WORKED SOLUTION

STEP III 2019, Question 7

Pure • Curve Sketching

STEP III 2019, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The equation contains only x^2 and y^2, so it is a quadratic in x^2 (or in y^2). That gives two tools at once: a discriminant, which tells us which values of y the curve can reach, and an explicit formula for x^2 in terms of y. For the shape near the origin we keep only the lowest-degree terms; far away we keep only the highest-degree terms. In the special case $a = b$ the equation factorises completely.

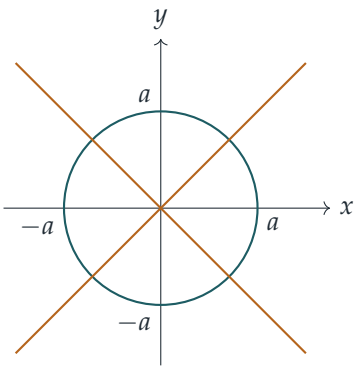
Solution

Part (i)

With $a = b$, move everything to one side and group the quartic and quadratic terms:

$$\begin{aligned} 0 &= (y^4 - x^4) - a^2(y^2 - x^2) \\ &= (y^2 - x^2)(y^2 + x^2) - a^2(y^2 - x^2) \\ &= (y^2 - x^2)(x^2 + y^2 - a^2). \end{aligned}$$

So a point lies on the curve exactly when $y = \pm x$ or $x^2 + y^2 = a^2$: the Devil's Curve is the pair of lines $y = x$ and $y = -x$ together with the circle of radius a centred at the origin.



Part (ii)(a)

Now $a = 2$, $b = \sqrt{5}$, and the curve is $y^4 - 5y^2 = x^4 - 4x^2$. Fix $y \geq 0$ and write $X = x^2$. A point (x, y) with $x \geq 0$ lies on the curve exactly when $X \geq 0$ satisfies

$$X^2 - 4X - (y^4 - 5y^2) = 0.$$

For this to have a real root, the discriminant must be non-negative:

$$\begin{aligned} 16 + 4(y^4 - 5y^2) &= 4(y^4 - 5y^2 + 4) = 4(y^2 - 1)(y^2 - 4) \\ &= 4(y+1)(y+2)(y-1)(y-2) \geq 0. \end{aligned}$$

Since $y \geq 0$, the factors $y + 1$ and $y + 2$ are positive, so we need $(y - 1)(y - 2) \geq 0$, that is,

$$0 \leq y \leq 1 \quad \text{or} \quad y \geq 2.$$

Conversely, every such y does occur. The roots are

$$X = 2 \pm \sqrt{D}, \qquad D = (y^2 - 1)(y^2 - 4),$$

and the larger one is at least 2, so it is a valid x^2. We will use this formula for x^2 below.

Part (ii)(b)

Close to the origin, the quartic terms y^4 and x^4 are tiny compared with y^2 and x^2. So the curve is approximately

$$-5y^2 \approx -4x^2, \qquad \text{that is} \qquad y \approx \tfrac{2}{\sqrt{5}}\, x \quad (x, y \geq 0).$$

The curve leaves the origin along the line of gradient $2/\sqrt{5} \approx 0.89$, a little below the line $y = x$. (The formula from (a) agrees: on the root $X = 2 - \sqrt{D}$, a binomial expansion gives $x^2 = \frac{5}{4}y^2 + O(y^4)$.)

Very far from the origin, the highest-degree terms dominate: $y^4 \approx x^4$, so $y \approx x$. We can do better. Write the equation as

$$(y^2 - x^2)(y^2 + x^2) = 5y^2 - 4x^2 \quad \Longrightarrow \quad y^2 - x^2 = \frac{5y^2 - 4x^2}{y^2 + x^2}.$$

Since $y/x \to 1$, the right-hand side tends to $\frac{5-4}{2} = \frac{1}{2}$. So $y^2 \approx x^2 + \frac{1}{2}$, which gives

$$y \approx x + \frac{1}{4x}.$$

The curve has the line $y = x$ as an asymptote and approaches it from *above*. In fact it never meets that line away from the origin: putting $y = x$ in the equation gives $x^2 = 0$.

Part (ii)(c)

Differentiating $y^4 - 5y^2 = x^4 - 4x^2$ implicitly with respect to x,

$$(4y^3 - 10y)\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 8x \qquad \Longrightarrow \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2x(x^2-2)}{y(2y^2-5)}.$$

Tangent parallel to the x-axis: we need the numerator to be zero and the denominator not. So $x = 0$ or $x = \sqrt{2}$.

- $x = 0$: then $y^4 = 5y^2$, so $y = 0$ or $y = \sqrt{5}$. At $(0, \sqrt{5})$ the denominator is $\sqrt{5}(10-5) \neq 0$, so this point counts.
- $x = \sqrt{2}$: then $y^4 - 5y^2 = 4 - 8$, so $y^4 - 5y^2 + 4 = 0$, which gives $y = 1$ or $y = 2$. The denominators there are -3 and 6, both non-zero.

? **Ask yourself** Why don't we count the origin, where the numerator is also zero?

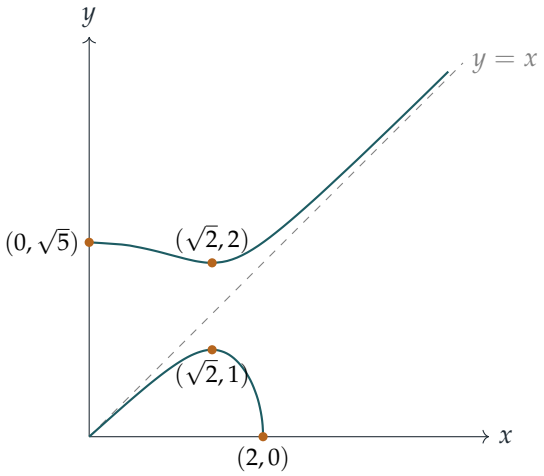
At $(0,0)$ the denominator vanishes too, so the formula gives $0/0$ and tells us nothing. Part (b) has already settled the question: the curve passes through the origin with gradient $2/\sqrt{5}$, not 0.

So the tangent is horizontal at $(0, \sqrt{5})$, $(\sqrt{2}, 1)$ and $(\sqrt{2}, 2)$.

Tangent parallel to the y-axis: we need the denominator to be zero and the numerator not, so $y = 0$ or $y^2 = \frac{5}{2}$. But $\sqrt{5/2} \approx 1.58$ lies strictly between 1 and 2, which part (a) rules out. With $y = 0$ we get $x^4 = 4x^2$, so $x = 0$ (the origin again, excluded as above) or $x = 2$. At $(2,0)$ the numerator is $2 \cdot 2 \cdot 2 = 8 \neq 0$. So the only such point is $(2,0)$.

The sketch for $x \geq 0, y \geq 0$ Using $x^2 = 2 \pm \sqrt{D}$, there are two pieces.

- **A loop for $0 \leq y \leq 1$.** The root $2 - \sqrt{D}$ runs from 0 (at $y = 0$) up to 2 (at $y = 1$), and $2 + \sqrt{D}$ runs from 4 down to 2. So the curve leaves the origin with gradient $2/\sqrt{5}$, rises to a maximum at $(\sqrt{2}, 1)$ and comes down to meet the x-axis vertically at $(2,0)$.
- **A wing for $y \geq 2$.** When $2 \leq y \leq \sqrt{5}$, the root $2 - \sqrt{D}$ runs from 2 down to 0; beyond $y = \sqrt{5}$ it is negative, so it is not a valid x^2. The root $2 + \sqrt{D}$ increases from 2 without bound. So this branch starts at $(0, \sqrt{5})$ with a horizontal tangent, dips to a minimum at $(\sqrt{2}, 2)$, and then rises, approaching $y = x$ from above.

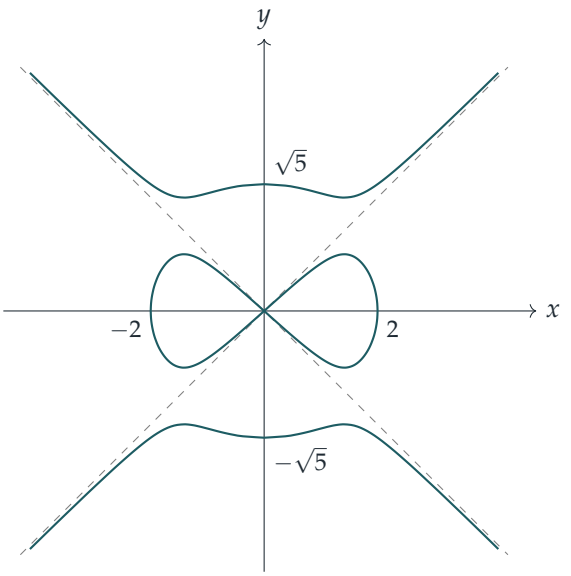


Part (iii)

The equation involves only x^2 and y^2, so the curve is symmetric in the y-axis (replace x by $-x$) and in the x-axis (replace y by $-y$). Reflecting the quarter-curve in both axes:

- The loop becomes a figure-of-eight between $x = -2$ and $x = 2$, with $|y| \leq 1$. Its two lobes cross at the origin along the lines $y = \pm\frac{2}{\sqrt{5}}x$, with turning points at $(\pm\sqrt{2}, \pm 1)$.
- The wing becomes two branches, one with $y \geq 2$ and one with $y \leq -2$. Each has a local maximum or minimum at $(0, \pm\sqrt{5})$ and turning points at $(\pm\sqrt{2}, \pm 2)$, and both approach the asymptotes $y = \pm x$ from outside.

No part of the curve lies in the strip $1 < |y| < 2$.



Checks

- Each marked point satisfies the equation. For example, at $(\sqrt{2}, 2)$ both sides equal -4, and at $(0, \sqrt{5})$ both sides equal 0.
- The point $(1, 1.5)$ is in the forbidden strip. There the left-hand side is $2.25 \times (-2.75) \approx -6.19$, while the right-hand side $x^2(x^2-4)$ can never be less than -4 (its minimum, at $x^2 = 2$). This is the discriminant argument in miniature.
- On the wing at $y = 100$, the formula gives $x^2 = 2 + \sqrt{(10^4-1)(10^4-4)}$. Then $4x(y-x) \approx 1.0002$, matching $y - x \approx \frac{1}{4x}$. Near the origin, the ratio y/x on the loop tends to $0.8944 \approx 2/\sqrt{5}$.

Common slips

- Listing $(0, 0)$ as a point with a horizontal (or vertical) tangent. There $\mathrm{d}y/\mathrm{d}x$ is $0/0$, and the true gradient comes from (b).
- Drawing the loop leaving the origin along $y = x$, or the wing crossing $y = x$. The loop starts at gradient $2/\sqrt{5} < 1$, and the wing stays strictly above $y = x$.
- Forgetting the condition $x^2 \geq 0$, which cuts the inner root off at $y = \sqrt{5}$.

Beyond STEP

The Devil's Curve is an *algebraic curve*: the zero set of a polynomial $F(x, y)$. The origin is a *singular point*, where both partial derivatives F_x and F_y vanish. The lowest-degree terms of F give the tangent lines there, which is exactly what we did in (b), and the

highest-degree terms give the directions of the asymptotes. At university these ideas reappear in the implicit function theorem and in algebraic geometry. Siklos, *Advanced Problems in Mathematics*, has more STEP-style curve-sketching problems with careful commentary.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.